Qi Zheng (郑启)

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Education

Tongji University — PhD in Computer Science and Technology (Sep 2022 – Jul 2026 (expected))

Research Focus: Spatial-Temporal Data Mining and Forecasting

Published 7 papers in related areas (4 as first author, including CCF-A/B conferences and IEEE Transactions TOP journals); 3 papers under review (including 1 first-author submission to ICLR 2026).

Tongji University — Master in Computer Science and Technology (Sep 2019 – Jun 2022) GPA Ranking: 3 / 53

Tongji University — Bachelor in Computer Science and Technology (Sep 2015 – Jun 2019) GPA Ranking: Top 10%

Selected Research Achievements

1. Towards Expanding-Node Spatial-Temporal Forecasting: A Structured Node Interaction Prompting Perspective.
ICLR 2026 (submitted, first author)

- *Problem & Method:* Existing forecasting models are tightly coupled with node parameters, limiting adaptation to expanding or removed nodes. Proposed a structured node-interaction prompting framework that constructs priors and dynamically refines them; new nodes are adapted through similarity-weighted few-shot initialization.
- *Results:* Compared with the SOTA model (STEV, 2025), SNIP reduces MAE by 3.71% and decreases GPU usage/training time by 92.5% / 91.3%, achieving a fully parameter–node-decoupled spatial-temporal framework for efficient scalable forecasting.

2. TLAST: A Time-Lag Aware Spatial-Temporal Transformer for Traffic Flow Forecasting.
IEEE TITS 2025 (first author, IEEE Transactions on Intelligent Transportation Systems, TOP Journal)

- *Problem & Method:* Transformer-based models ignore propagation time-lags and have quadratic computational cost. Introduced Time-Lag Embedding to model asynchronous correlations and Spatial Proxy Attention (SPA) to approximate global attention with proxy nodes, reducing complexity to O(N).
- *Results:* Compared with the SOTA model (STAEformer), TLAST improves MAE by 2.8% and reduces GPU usage / training time by 79.16% / 59.66%. Achieved a lag-aware Transformer architecture balancing high accuracy and linear efficiency for large-scale networks.

3. ST-ReP: Learning Predictive Representations Efficiently for Spatial-Temporal Forecasting. AAAI 2025 (first author)

- *Problem & Method:* Prior self-supervised spatial-temporal methods lack predictive representations and are hard to scale due to complex encoders. We proposed ST-ReP, a compact self-supervised framework with a “reconstruction + future prediction” objective and a linear-complexity “Compression–Extraction–Decompression (C-E-D)” encoder to directly learn predictive, lightweight representations.
- *Results:* Compared with the SOTA method (GPT-ST), ST-ReP reduces MAE by up to 20.24% and cuts GPU usage / training time by 67.22% / 28.07%. Developed a predictive and efficient representation learning framework for large-scale self-supervised spatial-temporal pretraining.

Research Projects

Ph.D. Dissertation (2022-2026) ***Efficient and Scalable Modeling of Dynamic Spatial-Temporal Data in Complex Scenarios***

Focuses on spatial-temporal data (e.g., traffic, climate, and energy sensor networks). Aims to build forecasting models that are high-performance, adaptive, and computationally efficient under dynamic temporal and spatial structures. Topics: spatial-temporal data mining, multivariate time-series analysis, self-supervised representation learning, and spatial-temporal foundation-model. Produced 7 papers (4 first author in AAAI, T-ITS, DASFAA, T-Big Data) and 3 under review (1 first author at ICLR 2026).

National Natural Science Foundation of China (NSFC) Project (2024 – 2027) ***Organizational Learning and Innovation Ecosystem with LLM Integration*** *(Cross-disciplinary research between Management Science and Computer Science)*

Participating as a core researcher in an interdisciplinary project bridging Management Science and Artificial Intelligence. Responsible for developing the LLMs and machine-learning framework to model enterprise innovation ecosystems and support data-driven policy generation, establishing a new empirical paradigm for management and economics research. This represents a first-of-its-kind initiative to apply LLM-based methodologies within management and economics research.

IKCEST Belt & Road International Big Data Competition (2020) ***Forecasting the Spread of High-Pathogenic Infectious Diseases***

Jointly organized by UNESCO IKCEST, the Chinese Academy of Engineering, Baidu, and Xi'an Jiaotong University.
The team analyzed data from 900+ regions across 11 cities to predict infectious-disease transmission trends. Qi was responsible for data analysis, feature engineering, and deep learning modeling, and also led the final project presentation and defense on behalf of the team. The team ranked 5th among 3,023 participants, winning the International Second Prize.

Skills

- *Programming:* Python, C++, JavaScript, SQL
- *Languages:* CET-6 (Good English reading and academic writing skills)
- *Expertise:* Deep Learning and Machine Learning methods for spatial-temporal data; experience in forecasting tasks and cross-disciplinary research integration.

Honors & Awards

- Outstanding Master Thesis of Tongji University (2023)
- 2nd Prize, IKCEST Belt & Road International Big Data Competition, Rank 5 / 3023 (2020)
- 2nd Prize, Chinese National Undergraduate Computer Design Competition (2017)
- Outstanding Student Scholarship, Tongji University (2016–2019, multiple years)

Self-Evaluation

Highly self-driven and goal-oriented with strong execution and learning ability. Open-minded and collaborative with solid research rigor and attention to detail. Enthusiastic about applying cutting-edge AI techniques to real-world spatial-temporal systems.

Publications (* indicates corresponding author)

1. **Qi Zheng,** Yao Zihao, Canyang Zhang, and Yaying Zhang*. Towards Expanding-Node Spatial-Temporal Forecasting: A Structured Node Interaction Prompting Perspective [C]// *International Conference on Learning Representations,* ICLR 2026. (Under review, Top-tier ML conference)
2. **Qi Zheng**, Minhua Shao, and Yaying Zhang*. TLAST: A Time-Lag Aware Spatial-Temporal Transformer for Traffic Flow Forecasting [J]. *IEEE Transactions on Intelligent Transportation Systems*, 26(9), 13144-13158, 2025, doi: 10.1109/TITS.2025.3583391. (IEEE Transactions, TOP journal)
3. **Qi Zheng,** Yao Zihao, and Yaying Zhang*. ST-ReP: Learning Predictive Representations Efficiently for Spatial-Temporal Forecasting [C]//*Proceedings of the AAAI Conference on Artificial Intelligence*, 39(12), 13419-13427, 2025, doi: https://doi.org/10.1609/aaai.v39i12.33465. (CCF-A, Top-tier AI conference)
4. Zhinan Xie, **Qi Zheng*** and Yaying Zhang, Temporal MLP Bridges the Gap Between Embedding and Attention for Multivariate Time Series Forecasting [C]//*2024 IEEE International Conference on Systems, Man, and Cybernetics (SMC),* Kuching, Malaysia, 2024, pp. 2373-2378, doi: 10.1109/SMC54092.2024.10831557.
5. **Qi Zheng** and Yaying Zhang*. TAGnn: Time Adjoint Graph Neural Network for Traffic Forecasting [C]//*Database Systems for Advanced Applications.* DASFAA 2023. Lecture Notes in Computer Science, vol 13943. 2023, doi: https://doi.org/10.1007/978-3-031-30637-2_24 (CCF-B Conference)
6. Canyang Zhang, **Qi Zheng,** and Yaying Zhang*. Spatial-Temporal Flow Holistic Interaction Graph Convolution Network for Bidirectional Traffic Flow Forecasting [C]// *2023 IEEE International Conference on Big Data (BigData),* Sorrento, Italy, 2023, pp. 1262-1268, doi: 10.1109/BigData59044.2023.10386746.
7. **Qi Zheng** and Yaying Zhang*. DSTAGCN: Dynamic Spatial-Temporal Adjacent Graph Convolutional Network for Traffic Forecasting [J]. *IEEE Transactions on Big Data*, 9(1), 241-253, 2023, doi: 10.1109/TBDATA.2022.3156366. (IEEE Transactions journal)
8. Huiyun Yu, **Qi Zheng**, Shuyun Qian and Yaying Zhang*. A Fuzzy-based Convolutional LSTM Network Approach for Citywide Traffic Flow Prediction [C]//*2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)*, Macau, China, 2022, pp. 3360-3367, doi: 10.1109/ITSC55140.2022.9922491.
9. Zihao Yao, **Qi Zheng**, Jiankai Zuo, and Yaying Zhang. GPS-Mamba: Graph Permutation Scanning State Space Model for Multivariate Time Series Forecasting [J]. *Expert Systems With Applications,* 2025. (under review)
10. Canyang Zhang, **Qi Zheng**, Aoyu Liu, Xiangmei Li, and Yaying Zhang. A Heterogeneity-Aware Spatial-Temporal Traffic Forecasting Model for Cross-City Generalization [J]. *Expert Systems With Applications,* 2025. (under review)